

LAB ASSIGNMENT: SSIS Data Flow Transformations

Objective:

Learn and implement various SSIS data flow transformations by building ETL packages that extract, transform, and load data using different transformation components.

Prerequisites:

- SQL Server Data Tools (SSDT) or Visual Studio with SSIS installed
- Access to SQL Server database with *Sales.Customer* and *Pricing* tables
- Basic understanding of SSIS data flow components

Lab Tasks:

Task 1: Review Demonstration

In class, we implemented an ETL package with:

- **Source:** ADO.NET Source (Connection 1)
- **Transformations:** Character Map and Aggregate transformations
- **Destination:** ADO.NET Destination (Connection 2)
- **Tables:** Sales.Customer and Pricing tables

Task 2: Explore SSIS Transformations

Visit the Microsoft documentation at: [SSIS Transformations Reference](#)

Study at least 5-6 different transformation types and understand their purposes.

Task 3: Implement New Transformations

Create a new SSIS package that includes **at least 2-3 different transformations** (other than Character Map and Aggregate). Use the same source tables (Sales.Customer and Pricing) or similar tables from your database.

Suggested transformations to explore: Derived Column, Conditional Split, Lookup, Merge Join, Data Conversion, Sort, Union All, Multicast

Deliverables:

- 1 **SSIS Package (.dtsx file)** - Your complete SSIS solution
- 2 **Screenshots** - Data flow design showing all transformations with connections
- 3 **Documentation** - Brief explanation (1-2 paragraphs) for each transformation used: purpose, configuration, and output
- 4 **Sample Output** - Screenshots of transformed data or destination table results

Submission Guidelines:

Submit all deliverables in a single ZIP file named **YourName_SSIS_Lab.zip** through the course portal by the assigned deadline.

Note: Focus on understanding how each transformation processes data. Experiment with different configurations and observe the results.